

This Month's DVD Contents

This month's DVD brings to you the design and simulation software from National Instruments as well as DesignSpark PCB, along with Android SDK to help you develop your own Android app. Also included are some popular resources

DILIN ANAND

NI Circuit Design Suite

National Instruments' Circuit Design Suite 13.0.1 Education Edition includes NI Multisim 13.0 and NI Ultiboard 13.0. Multisim allows advanced mixed-mode simulation and validation with a library of components to easily build and validate circuit behaviour. Ultiboard is a flexible PCB layout and routing tool which also allows one to transfer the complete Multisim schematics to Ultiboard and layout, and then to route and export the completed designs for fabrication.

GCC ARM Embedded

GNU Compiler Collection (GCC) ARM Embedded contains a pre-built GNU toolchain for ARM Cortex-R (Cortex-R4/R5) and Cortex-M (Cortex-M0/M0+/M3/M4) processors. It includes front ends for C, C++, Objective-C, Fortran, Java, Ada and Go, as well as libraries for these languages (libstdc++, libgccj,...).

DesignSpark PCB

DesignSpark PCB design software can significantly reduce your concept-to-production time. Its powerful software engine enables one to capture schematics and design PCB boards and layouts. It also

provides the simulation tool (LTSpice, Tina, etc) interface facility. Its built-in design calculator (track/via resistance, power dissipation, etc) enhances its usability.

DesignSpark Mechanical

DesignSpark Mechanical helps quickly design and change product concepts in 3D without having to learn complex traditional CAD software. It helps in producing a highly detailed dimensioned worksheet, remove bottlenecks, insert free, off-the-shelf component 3D models and create geometries with powerful and intuitive gesture based modelling.

KTechlab

KTechlab allows one to design and simulate the circuits related to PIC microcontroller and other electronics. It is an integrated environment for electronic design and automation (EDA) tasks. KTechlab features an extensive circuit designer that helps autorouting along with the simulation of logic elements and common electronic components.

Printed Circuit Board

Popularly known as PCB, it is an interactive printed circuit board editor that is used for all types of systems like Unix,

Linux, Windows and Mac. It can produce industry standard RS-274X, Excellon NC-Drill and centroid data (X-Y data) output format for use in board fabrication. It also provides photo-realistic and design-review images.

SZARP SCADA

SZARP is a full-featured, open source (GPL) SCADA system for Linux to perform human-machine interface visualisation tasks. Its latest version is compatible with 32-bit MS Windows (NT/2000/XP) and Linux. It has been mostly written in C++ with an average number of source code comments.

Engineering Calculator

Engineering Calculator is a lightweight yet handy math expression evaluation and unit conversion utility. Its portability allows the installation process to be bypassed easily and to be used on any PC by simply moving or copying the program files to a USB thumb drive or other storage device. The best part is that it will not update the Windows registry or hard drive with new entries without approval, and will leave no traces of it on the disk once removed.

Android SDK

Android SDK is a software stack for mobile devices, including an operating system, middleware and key applications, to help build, test and debug apps. If used properly, the SDK, together with Eclipse (the officially supported IDE) and JDK (Java Development Kit), can be state-of-the-art software for Android devices. Android SDK assumes that developers already have the Eclipse IDE installed, as well as JDK and the Android Development Tools plugin (not compulsory, although recommended,) as it includes only the basic utilities for Android app development. ●

Popular Resources

1. **7Zip.** 7Zip is an open source tool for compressing and decompressing files.
2. **JRE.** Java Runtime Environment, version 7 is used to run already developed Java programs.
3. **VLC Media Player.** VLC Media Player, version 2.1.3 is a highly portable multimedia player for various audio and video formats (MPEG-1, MPEG-2, MPEG-4, DivX, MP3, ogg, 3gp, etc) as well as DVDs, VCDs and various streaming protocols.
4. **Opera.** Opera web browser, version 23.0.1522.60 (latest version), developed by Opera Software runs on Microsoft Windows and OS X operating systems and uses the Blink layout engine.
5. **Free Download Manager.** Free Download Manager is an open source downloader which can increase download speeds up to 600%.
6. **Thunderbird.** Thunderbird, version 24.6.0 is a free email application that's easy to set up and customise, and is loaded with great features.
7. **Mozilla Firefox.** Mozilla Firefox, version 31.0b8 is a light and tidy open source web browser with simple and effective UI, fast speed and strong security capabilities.
8. **Apache OpenOffice.** Apache OpenOffice, version 4.1.0 is the leading open source office software suite for word processing, spreadsheets, presentations, graphics, databases and more.
9. **Protector Plus Internet Security.** Protector Plus antivirus software protects your computer against all types of malware like viruses, Trojans, worms and spyware.